DustWise: Water-Aware Solar Panel Cleaning in the Thar Desert with Reinforcement Learning

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GitHub: `https://github.com/RamcharanRajpurohit/dustwise`

1 Problem

Desert dust cuts the output of Rajasthan's solar plants by up to 30%, but cleaning costs labour and, for a water wash, scarce trucked water. Light rain makes things worse: it cements loose dust so that only water removes it, while heavy rain cleans panels for free. A plant operator must decide every day which part of the plant to clean and how. We model a 10 MWp plant in Jaisalmer as 10 blocks of 1 MWp (1,820 modules each), with one cleaning crew that treats one block per day, and train an RL agent to make that daily decision. The environment is driven by real data for Jaisalmer (26.91 N, 70.91 E), 2022-08 to 2025: ERA5 irradiance and rain, CAMS dust, and archived day-ahead forecasts, all from Open-Meteo.

2 RL formulation

State $\mathcal{S}$ (26-dim, continuous). For each block i: loose soiling loss ℓ_i and cemented loss c_i (scaled $\times 10$); tank level $w/10^4$ L; the day-ahead forecast of irradiance $\hat{H}_t/7$ and rain $\min(\hat{R}_t, 12)/6$; today's CAMS dust $D_t/9500$; and the season, $(1+\sin\phi_t)/2$ and $(1+\cos\phi_t)/2$ with $\phi_t = 2\pi\,\mathrm{doy}/365.25$.

Action $\mathcal{A}$ (discrete, 21). $a = 0$ idle; $a = 1..10$ dry-brush block i; $a = 11..20$ water-wash block i. Washes are masked out when $w < 3640$ L (one wash).

Dynamics (one step = one day). In the morning a brush sets $\ell_i \leftarrow 0.1\,\ell_i$; a wash sets $\ell_i = c_i = 0$ and $w \leftarrow w - 3640$. The plant then generates with the *actual* irradiance H_t. Afterwards the actual rain R_t updates the dust: if $R_t \geq 6$ mm all blocks are clean; if $0 < R_t < 6$, $c_i \leftarrow c_i + \ell_i$, $\ell_i \leftarrow 0$. New dust settles as $\ell_i \leftarrow \ell_i + 0.004\,D_t/\bar{D}$ ($\bar{D} = 950\,\mu\mathrm{g/m^3}$), and the tank refills by 1,000 L/day up to 10,000 L.

Reward $\mathcal{R}$. The negative daily cost in units of 10^4 INR:

$$r_t = -10^{-4}\Big(\tau P\,\mathrm{PR}\,H_t \textstyle\sum_i \min(\ell_i + c_i,\, 0.3) + 364\,[a \neq 0] + 874\,[a > 10]\Big)$$

where the first term is lost revenue and the others labour and water, tariff $\tau = 2.5$ INR/kWh, $P = 1000$ kWp and PR $= 0.78$. Labour is 0.20 INR and water 2 L at 0.24 INR/L per module. The agent therefore trades certain cleaning costs now against uncertain soiling losses that may be erased by rain.

Constants come from published sources: soiling 0.37%/day measured in Gandhinagar [1], the 6 mm rain threshold and 30% cap of the Kimber model [3], rain cementation [1, 2], and Indian tender and auction prices for labour, water and tariff (listed in the README).

Episodes are 90 random consecutive days from 2022-08 to 2024. 2025 is the held-out test year, played as one 364-day episode. Archived forecasts exist only from 2024-01-20, so earlier training days use the observed weather as the forecast; on 2024–25 the real forecast has irradiance correlation 0.89 (nRMSE 13%) and caught only 14 of 25 heavy-rain days with 18 false alarms. The agent must act under genuine forecast uncertainty.

3 Algorithm and baselines

Baselines. No cleaning; daily rotation with a brush; daily rotation with a wash; and *greedy*: brush the dustiest block unless heavy rain is forecast.

Algorithm. MaskablePPO [7] from Stable-Baselines3 [6], with a 128×128 MLP, $\gamma = 0.9$ (soiling effects decay within days) and an entropy bonus of 0.01, trained for 1M steps over 3 seeds. The policy receives the tank mask, so it never wastes a crew-day on an impossible wash. PPO's clipped on-policy updates were more stable here than value-based methods (Section 5).

4 Results on unseen 2025

Policy	Cost (lakh INR)	Lost MWh	Water kL
No cleaning	47.79	1912	0
Rotation wash	25.29	961	371
Rotation brush	19.62	732	0
Greedy brush	19.63	735	0
MaskablePPO	**13.00±0.04**	**467**	**175**

Table 1: Total 2025 cost with real forecasts; the agent is mean ± std over 3 seeds.

Main result. The agent costs 13.00 lakh INR on 2025, **34% less than the best baseline**, with 36% less energy lost and less than half the water of the wash rotation. Its 3 seeds agree within 0.04.

What it learned. Unlike greedy, the agent spends about 111 days idle, 205 brushing and 48 washing. 96% of its washes go to blocks holding cemented dust, the only case a brush cannot fix, so water is used where it is irreplaceable.

5 What we tried, in order

1. **DQN with perfect forecasts** (observed weather as forecast): 18.01 vs. greedy 19.41.
2. **Real day-ahead forecasts**, then five algorithms, 3 seeds, 500k steps: DQN 17.30, QR-DQN 18.80, PPO 15.39, A2C 16.22. On-policy methods were better and more stable.
3. **Action masking (MaskablePPO): 14.18.** Without masks DQN wasted crew-days on washes with an empty tank, badly so when water was scarce.
4. **Tuning on a 2024 validation year** (training on 2022–23). The season input and entropy bonus helped most (2024 cost 19.16→17.55); PPO settings from An [4] and 2–3-day rain forecasts (as in Terhag et al. [5]) gained less and were dropped. This gives the final agent, 13.00.

Limitations. Same-day dust is observed, not forecast; training years before 2024 lack real forecasts; soiling constants come from nearby studies rather than

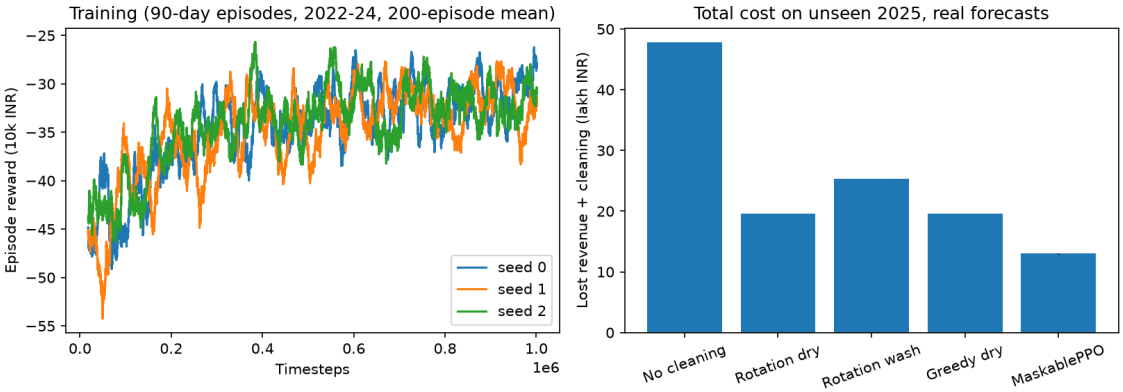


Figure 1: Left: training reward on 90-day episodes, 3 seeds. Right: total 2025 cost of each policy.

Jaisalmer measurements; and MaskablePPO itself was picked on the 2025 comparison, though the final additions were chosen on 2024 only.

References

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[2] D. Olivares et al., "Soiling in the Atacama Desert: impacts on solar performance and evaluation of cleaning techniques," *Renewable Energy* 257, 2026. doi:10.1016/j.renene.2025.124815

[3] pvlib-python, Kimber soiling model, `https://pvlib-python.readthedocs.io`

[4] H. An, "Reinforcement learning-based dynamic cleaning scheduling framework for solar energy system," arXiv:2603.07518, 2026.

[5] F. Terhag et al., "Optimization of cleaning strategies based on ANN algorithms assessing the benefit of soiling rate forecasts," *AIP Conf. Proc.* 2126, 220005, 2019.

[6] A. Raffin et al., "Stable-Baselines3: reliable reinforcement learning implementations," *JMLR* 22(268), 2021.

[7] S. Huang and S. Ontañón, "A closer look at invalid action masking in policy gradient algorithms," *FLAIRS*, 2022.